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Numerical simulation of droplet deposition onto a liquid film by VOF–MPS hybrid method

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Abstract Droplet entrainment and deposition are a couple of significant mechanisms for the heat transfer in annular two-phase flows existing in some heat exchange systems. The basic physics include the peeling of droplets from the liquid film due to high friction with the gas phase and the collision of droplets with the liquid film or deposition into the liquid film. Droplet deposition particularly plays a crucial role in the course of film dryout events which might have a vital importance for particular systems. In this study, a new numerical method (named as VOF–MPS hybrid method) based on the moving particle semi-implicit (MPS) method was developed to analyze the droplet deposition onto a stagnant thin liquid film. That proposed method combines the volume of fluid (VOF) solver of the open-source CFD code OpenFOAM with the MPS method. VOF–MPS technique introduces the surface tension force calculation of VOF model into the MPS method. MPS method formerly employed the continuum surface force (CSF) approach based on the particles. Three-dimensional (3D) VOF–MPS simulation with the novel surface tension modeling addition to the MPS-based modeling provides a smoother liquid–gas interface on the crown formed after the impact. Droplet deposition experiments were also carried out for the validation and comparison of two models. VOF–MPS method could predict the crown parameters such as crown thickness relatively better than the MPS method employing the CSF. However, the instabilities formed at the tip of the crown, observed in the experiments, could not be resolved with both methods.

Keywords Particle method · VOF · Droplet deposition

1 Introduction

Droplet entrainment and deposition are important phenomena in two-phase gas–liquid flows existing such as in boiling water reactors (BWR) as well as other heat exchange systems in which boiling takes place. They are the mechanistic events in which liquid particles dispatch from the liquid film surface and deposited back onto the liquid film or onto the wall. Those events of entrainment and deposition may significantly influence the energy transfer between the gas core and the liquid film adjacent to the wall. Especially, the heat flux in

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film dryout conditions is highly affected by the amount of liquid particles entrained into the gas core flow. Hence, for instance, effectiveness of emergency core cooling systems of nuclear power plants, which floods the high temperature core structures with the water, is directly related to the liquid entrainment and deposition phenomena (Ishii and Mishima 1989).

The basic physics includes the peeling of droplets from the liquid film due to high friction on the interface and the collision of droplets with the liquid film followed by the deposition back into the liquid phase. Droplet deposition particularly might play a crucial role in the course of film dryout events. Therefore, accurate prediction of the deposition is important for the modeling of heat transfer. For the purpose of robust heat transfer estimations, several models have been proposed in the literature to investigate the entrainment and deposition phenomena. Ishii and Grolmes (1975) grouped the entrainment mechanism into five categories depending on the wavy structures on the liquid film surface. The prediction of these phenomena mainly relies on the empirical correlations. Some examples of them have been proposed in the literature by Paras and Karabelas (1991), Kataoka et al. (2000), and so on. Azzopardi (1997) comprehensively reviewed the proposed correlations related to both entrainment and deposition.

Droplet deposition rate is a function of the physical properties such as viscosity and densities of gas–liquid, surface tension of liquid phase, and droplet velocity (Nigmatulin et al. 1996). As Azzopardi (1997) reported in his review, larger drops do not move in a random manner but travel in straight lines due to the initial stimulus given to the drops by the entrainment process. According to the same review, it is stated that lateral motion of larger drops is unaffected from the turbulence while the motion of smaller ones appears more random. Therefore, the basic physics of droplet deposition onto liquid film can be studied by imitating a vertical collision of a droplet with the liquid film surface.

After the droplet impact, a crown structure formed and its properties such as height, thickness, and rate of spreading which depend mainly on the impact speed and liquid properties as well. Most of the researchers focused on such quantitative parameters to investigate the deposition phenomenon since the first attempt to model the droplet deposition onto a liquid film performed by Harlow and Shannon (1967). For instance, Yarin and Weiss (1995) derived a theoretical correlation for the spreading rate of the crown. Weiss and Yarin (1999) investigated that event numerically and they found a good agreement in spreading rate with the correlation of Yarin and Weiss (1995). Rieber and Frohn (1999) performed a simulation using the volume of fluid (VOF) method in 3D. They focused on the number of fingers appearing due to the Rayleigh instability. They found good agreement with a theoretical model.

Despite the numerous researches based on the empirical correlations and numerical calculations, investigating the fundamental mechanism with the use of different approaches is still important to improve the understanding of the droplet deposition further.

One of the distinct methods applied to the droplet deposition was the moving particle semi-implicit (MPS) numerical technique developed by Xie et al. (2004). They investigated the droplet collision and splashing on the moving liquid film. However, in that study, the resolution of particles was so coarse that precise shape of the crown interface could not be reproduced. Additionally, the interface boundary was hard to be determined precisely and small disturbances originated from the particle locations may have created instability in surface tension force calculation. Due to this fact, a smoothing mechanism has a potential to increase the stability of the computation. For that purpose, in this study, we developed a new numerical approach to investigate the water droplet deposition onto a thin water film with an improved surface tension modeling added to the MPS method. The novel numerical method, named as VOF–MPS hybrid method, integrates the MPS method into conventional volume of fluid (VOF) method. To combine both methods, MPS method is added to the open source CFD tool OpenFOAM. Liu and Koshizuka (2005) have already proposed a combined method of MPS and finite volume method for only two-dimensional (2D) simulation. Hereafter, their method is abbreviated as MPS–CSF and it is used for the comparisons with our results. However, surface tension modeling in their method the so-called continuum surface force (CSF) model was still based on the particles that may have not provided sufficient stability as it is stated in their report. In addition to the improved calculation of the surface tension force, the proposed model here advances the hybrid method to 3D numerical simulation capability. VOF–MPS method is also validated against the experimental data qualitatively and quantitatively together with the MPS–CSF method.

2 Numerical method

The VOF–MPS hybrid method is a combination of VOF method (for mesh calculation) (Hirt and Nichols 1981) and MPS method (for particle calculation) (Koshizuka and Oka 1996). For the mesh calculation, one

of the solvers in OpenFOAM, which is an open source CFD software (Weller et al. 1998), is utilized. The whole calculation domain is divided into Eulerian mesh. Some part of the computational domain is occupied by the particles representing the liquid phase. Figure 1 shows the flowchart of calculation algorithm. The calculation consists of three parts; mesh calculation part, particle calculation part and information transfer part. The fluid flow parameters including velocity and pressure are calculated for all computational cells, and the particles are moved according to the calculated velocity. Moving particles provide the advection of the density and the viscosity fields. Information transfer module provides the communication to couple the mesh and the particle calculation. According to the information in the particle, gas–liquid phase distribution and liquid volume fraction are interpolated to the mesh.

In the VOF method mass and momentum conservation equations for the gas–liquid two-phase flow are as follows:

$$\nabla \times u = 0 \quad (1)$$

$$\frac{\partial \rho u}{\partial t} + \nabla \times (\rho u u) = -\nabla p + \rho g + \nabla \times \mu(\nabla u + \nabla^t u) + F_\sigma \quad (2)$$

where u , ρ , p , g , μ , and F_σ are the velocity, density, pressure, gravitational acceleration, viscosity and surface tension force, respectively. Above equations are solved based on the viscosity and density obtained from the following formulas:

$$\rho = (1 - \alpha)\rho_g + \alpha\rho_l \quad (3)$$

$$\mu = (1 - \alpha)\mu_g + \alpha\mu_l \quad (4)$$

where indices g and l indicate gas and liquid phases, respectively. Liquid volume fraction α is defined in each discretized cells as $\alpha = 1$ in the liquid phase, $\alpha = 0$ in the gas phase and $0 < \alpha < 1$ on the interface. In the proposed hybrid method, density and viscosity are calculated by the liquid volume fraction extracted from the particle distribution.

Surface tension is calculated explicitly in the momentum equation as an external force. Surface tension coefficient is assumed to be uniform and steady, and the temperature dependence and surfactant effect are not considered. In the MPS–CSF model proposed by Liu and Koshizuka (2005), surface tension is calculated from the particles. However, this method may occasionally lead to large disturbances in the calculation due to the particle position deviations on the interface boundary. To avoid the disturbance and to improve the stability, instead of the particles, current method employs the mesh part for the surface tension calculation utilizing the following formulas implemented in the VOF solver of OpenFOAM (Brackbill et al. 1992):

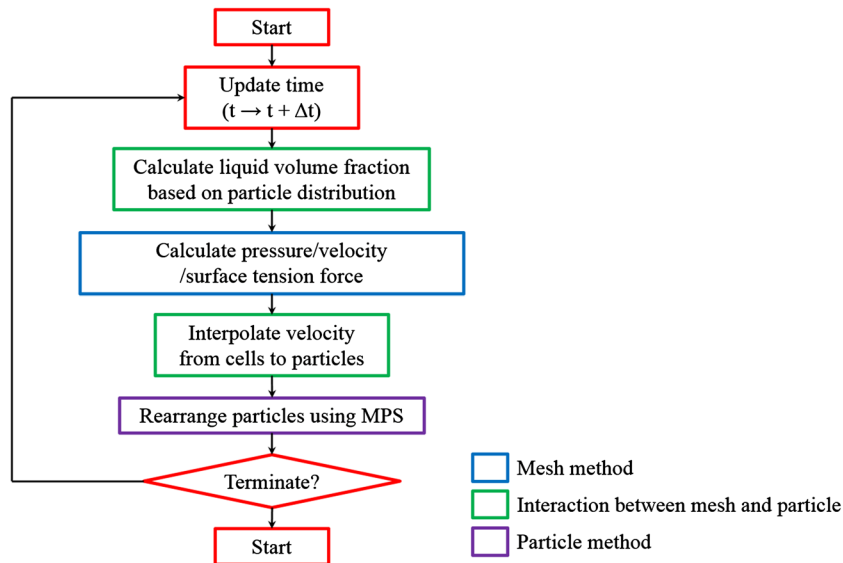


Fig. 1 Algorithm of VOF–MPS hybrid method

$$F_\sigma = \sigma \kappa \nabla \alpha \quad (5)$$

$$\kappa = -\nabla \times \left[\frac{\nabla \alpha}{|\nabla \alpha|} \right] \quad (6)$$

where σ denotes the surface tension and κ is the mean surface curvature. In this study, $\nabla \alpha$ is valid only at the interface region where volume fraction changes, and it is positive in the direction from gas to liquid phase.

In the VOF–MPS method, while the mesh is in Eulerian description, the particles are in Lagrangian description. Since the gas–liquid interface is defined by the particle distribution, information exchange between mesh and particle is required. The physical properties to be exchanged are the liquid volume on each particle and velocity vector on each mesh. These values are transferred to each other by volume-weighted averaging method.

Figure 2 shows the schematic representation of information exchange between the mesh part and particle part. Each particle is assumed to have volumes with a dimension of constant distance between the particles. Quantity of a physical property f carried by a particle is distributed to the surrounding meshes (to four cells in 2D, to eight cells in 3D) depending on the ratio of particle volume to the cell volume which is formulated as follows:

$$\langle f \rangle_k = \frac{1}{V} V_k f \quad (7)$$

V is the cell volume, V_k is the portion of a particle volume occupying in the surrounding k th cell. The information transfer from a mesh to particles is done in a similar way considering the volume ratios of the cells and particles.

To reduce the computational time, a quarter of a cube volume ($5.4 \times 6 \times 5.4$ mm) containing a quarter of the droplet was modeled with the periodic boundary conditions (Fig. 3). Dimensions of the computational domain were determined as the minimum values above which calculation results do not change significantly. This allowed us to perform the simulation with minimal computation time without any numerical disturbance which might be originated from the boundaries. The gravity was in the direction of negative y coordinate. Positive side of x and z coordinates was taken as computational domain. A rectangular computational domain was selected; this is because the information exchange module works with the rectangular cells as explained above.

Except the periodic boundaries and upper surface, the other surfaces were defined as the wall boundary with the conditions of the zero pressure gradient and zero velocity. Upper surface was taken as atmospheric constant pressure boundary.

3 Experimental setup

A schematic layout of the experimental setup for water droplet deposition onto water film is illustrated in Fig. 4. A syringe pump was used to inject the water through a vinyl tube at a stable flow rate for obtaining uniform droplet sizes. Droplets detached from the stainless-steel needle tip when the gravity overcame the

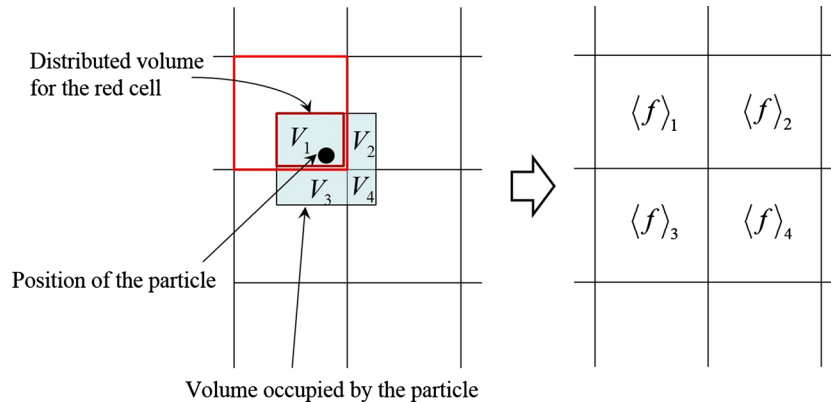


Fig. 2 Volume weighted averaging method for mesh-particle information exchange

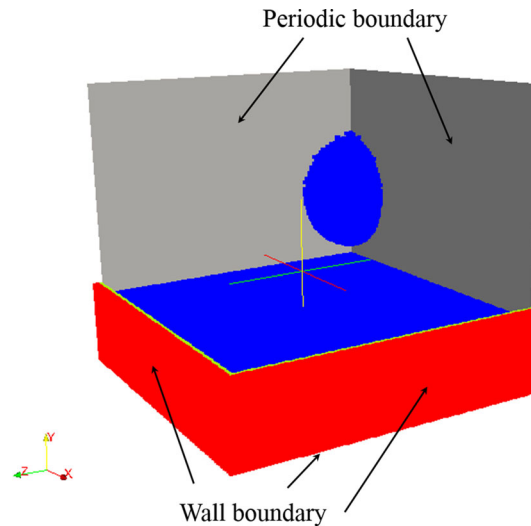


Fig. 3 Computational geometry for numerical simulation

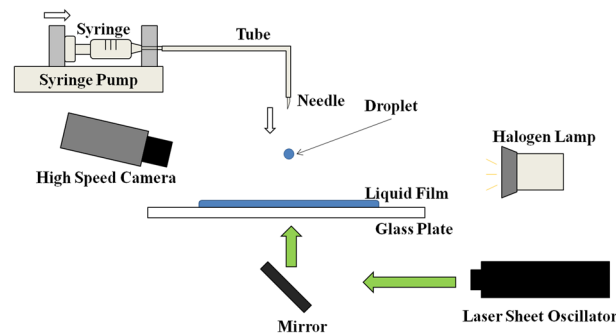


Fig. 4 Experimental setup

surface tension and impinge onto a 1-mm-thick water film which was created by spreading the water on a glass substrate. Outer boundaries of the film were stabilized with its surface tension due to a large contact angle which is around $70\text{--}80^\circ$. The increase in the thickness of the film due to the added mass by the impacting drops was balanced with the free spreading of liquid film. The time intervals between the droplet impacts were kept sufficiently long so that surface waves could be dumped and film thickness could be stabilized with lateral expansion of the film.

A high-speed camera was employed to record the impact process at high time resolutions. Two methods of illumination were used for the visualization of impact process and crown formation. A halogen lamp was employed as a light source for the background shadowgraphy. A CW green laser light sheet illumination was also used to obtain a clear view of the crown boundary with laser-induced fluorescence. For that purpose, the water film was dyed with the Rhodamine-B to visualize a sharp boundary with the help of fluorescence light. Laser light reflections were filtrated with a cutoff filter to eliminate the noises in the images Fig. 5.

Two sample images of droplet impacts having the impact speeds of 2.4 and 2.9 m/s are presented in Fig. 6 recorded with shadowgraphy and laser-induced fluorescence method. With the laser illumination, more contrasted and sharp cross-sectional image of the crown was obtained. The quantitative parameters, which could be resolved from the time sequential images of the crown, are shown in Fig. 6.

Droplet diameters, water film thickness, and impact velocities were measured from the shadowgraphy images prior to the impact. Equivalent droplet diameters and film thickness were extracted by image area thresholding method. Uncertainty of those type of measurements ranges in between 10 and 20 % (Yoon et al. 2004). Droplet impact velocities were calculated by center of mass tracking method utilizing the time sequential shadowgraphy images. Droplet center of mass was calculated based on its area which was also

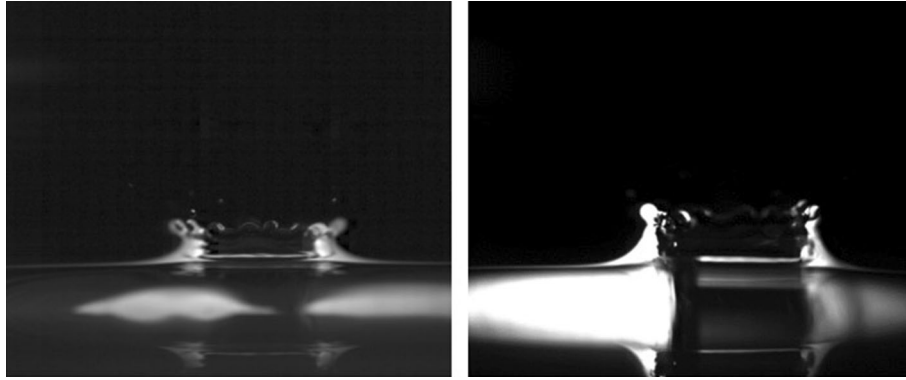


Fig. 5 Samples of shadowgraph (*left*) and fluorescence images (*right*)

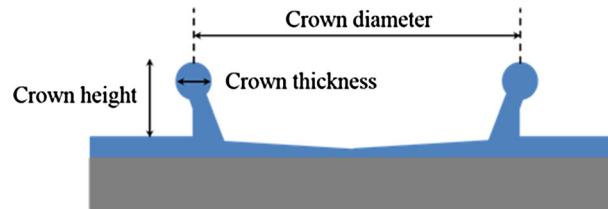


Fig. 6 Crown parameters

extracted by the image thresholding. The method is similar to the droplet diameter measurement, hence velocity measurement uncertainty has the similar range with the droplet diameter measurement.

It is a well-known fact that dissolved substances (surfactants) in the water or other type of liquids may reduce the surface tension (Eastoe and Dalton 2000). With this in mind, we investigated the surfactant effect of Rhodamine-B on some crown parameters with the preliminary tests performed using pure water and Rhodamine-B additive. Based on the obtained results, Rhodamine-B has negligible effect on the crown diameter and it has a small influence on the crown height which is still below the level of experimental uncertainty.

4 Results and discussions

4.1 Validation of VOF–MPS method

In this section, we performed a validation study with an experimental case of the water droplet deposition onto the water film. Samples of time sequential images and simulation results at the identical instants of time are presented in Fig. 7. The simulations were performed for two different surface tension coefficients in order to explore the sensitivity of the model against the surface tension coefficient. In the experiment, droplets having 2.5 mm in average diameter fell down from 50 cm height and impinged onto the water film (1.2 mm thick) at an impact speeds of 3.0 m/s. Water surface tensions were taken as 71.7×10^{-3} and 36×10^{-3} N/m, the latter of which corresponds to almost half of the pure water surface tension. As can be seen from Fig. 7, the crown shape preserves its similarity in three cases until 2 ms. After that time, for the reduced surface tension case, main body of the crown gets thinner and more elongated compared to the normal surface tension. Additionally, the neck formed between the rim and the main body of the crown is thinner and more discernable than the normal surface tension case.

Figure 8 shows the temporal variation of the crown parameters, which were normalized by initial droplet diameter. The time was non-dimensionalized with the ratio of droplet impact velocity and the diameter (tU/D). After the non-dimensional time of 3, extension of crown diameter is accelerated in the case of reduced surface tension (Fig. 8a). On the other hand, experimental results and normal surface tension case are relatively in good agreement until the non-dimensional time of 3. After that moment, a slight overestimation begins to appear.

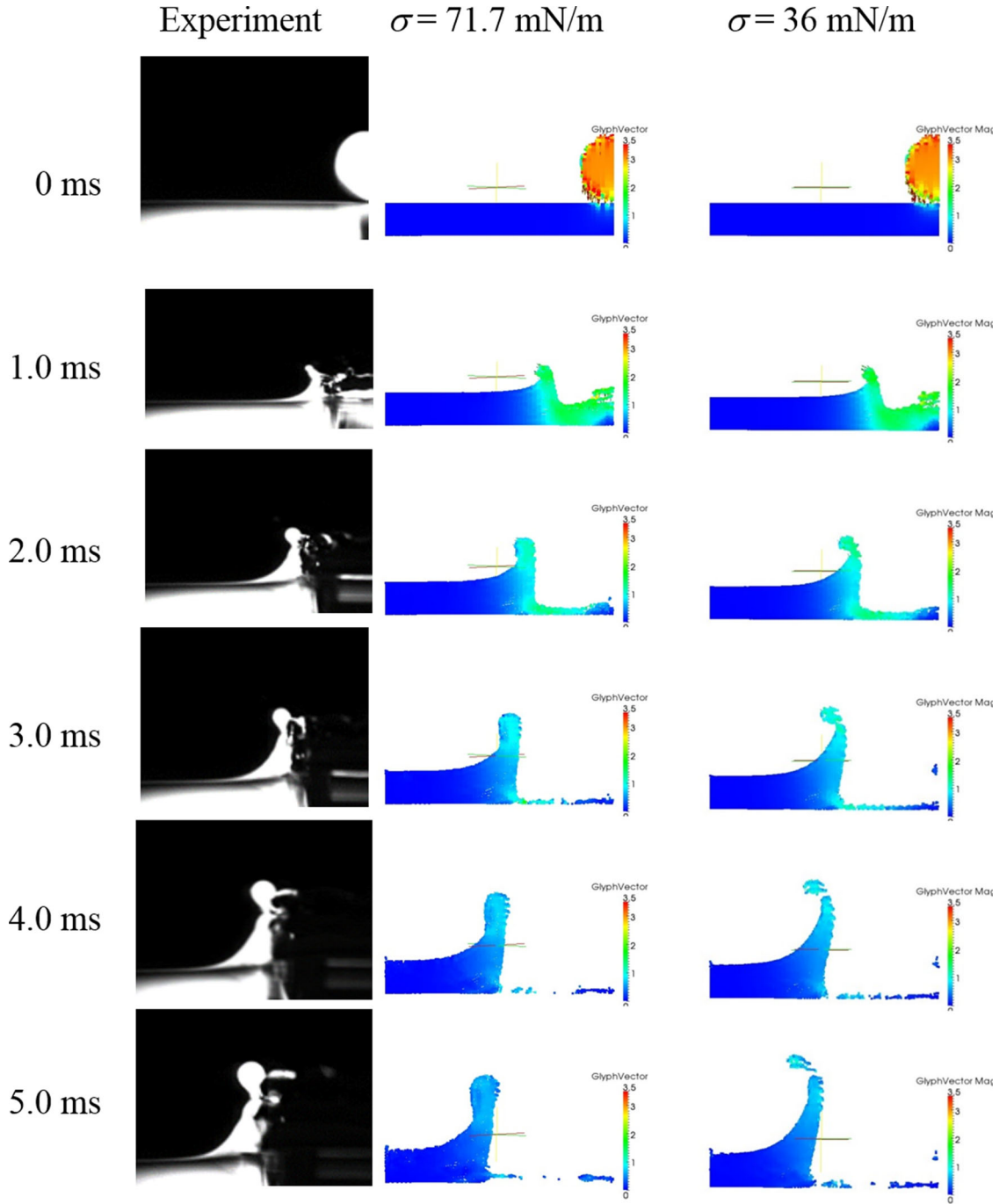


Fig. 7 Time sequential images of experimental and VOF-MPS simulation results. The particles are colored by the velocity magnitude. The impact velocity was 3.0 m/s

Temporal variations of crown heights are underestimated in both cases (Fig. 8b). The difference between the experimental results and normal surface tension case is larger compared to those of the reduced surface tension case. Despite the larger discrepancy, non-linear tendency of the simulation is consistent with the experiment. On the contrary, the crown height varies linearly in time for the reduced surface tension case. In the simulations, rim structure formed at the top of the hill (above the main body of the crown) has a smooth shape. On the other hand, in the experiment finger-like extrusions from the rim were observed due to the interface instabilities, even though they were not visualized from the experimental images in Fig. 8. Those

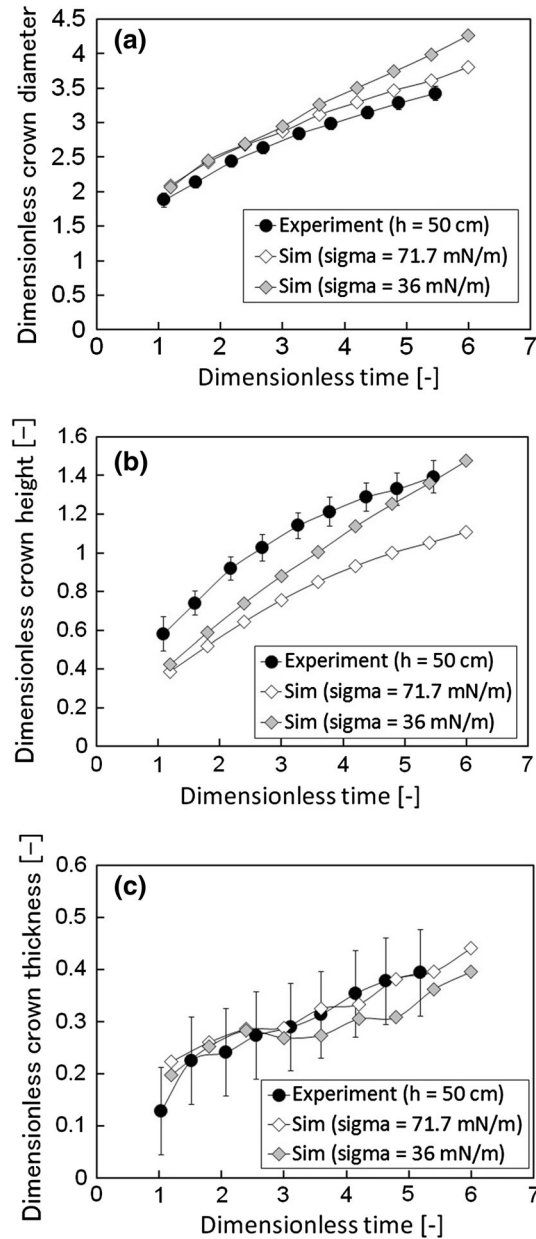


Fig. 8 Comparison of non-dimensional crown parameters. **a** Diameter, **b** height, **c** thickness

extrusions are likely to cause the experimental crown heights to be detected as higher than the main body of the crown or rim.

Prediction of the crown thickness has better agreement with the experimental results for both cases (Fig. 8c). The curve for the reduced surface tension case starts to diverge from the experimental results after around the non-dimensional time of 3. Nevertheless, it is still in the range of experimental deviations.

4.2 Comparison of VOF–MPS and MPS–CSF models

We performed a comparative analysis of our model (VOF–MPS) with the particle method (MPS–CSF) from the aspects of crown parameters and shapes. For that purpose, we used the same experimental results whose parameters were already given above. This time, MPS–CSF method is tested with the reduced surface tension which has the value of 36×10^{-3} N/m.

Time sequential instances of the impact process were illustrated in Fig. 9 for all cases. The results for the VOF-MPS and MPS-CSF demonstrate similar tendencies except that the crown is relatively more inclined towards the center for the case of MPS-CSF. In the case of reduced surface tension (denoted as half sigma in Fig. 9) a large size of the rim formation at the top is not observed since the lower surface tension leads to the liquid breakup from the tip easily.

In both cases of MPS-CSF method, the liquid interface is not smooth and more indented due to the particle-based surface tension calculation feature. Interface irregularities are concentrated on the inner side of the crown especially at the instants of 5.0 and 4.0 ms. As it was previously mentioned, the uneven

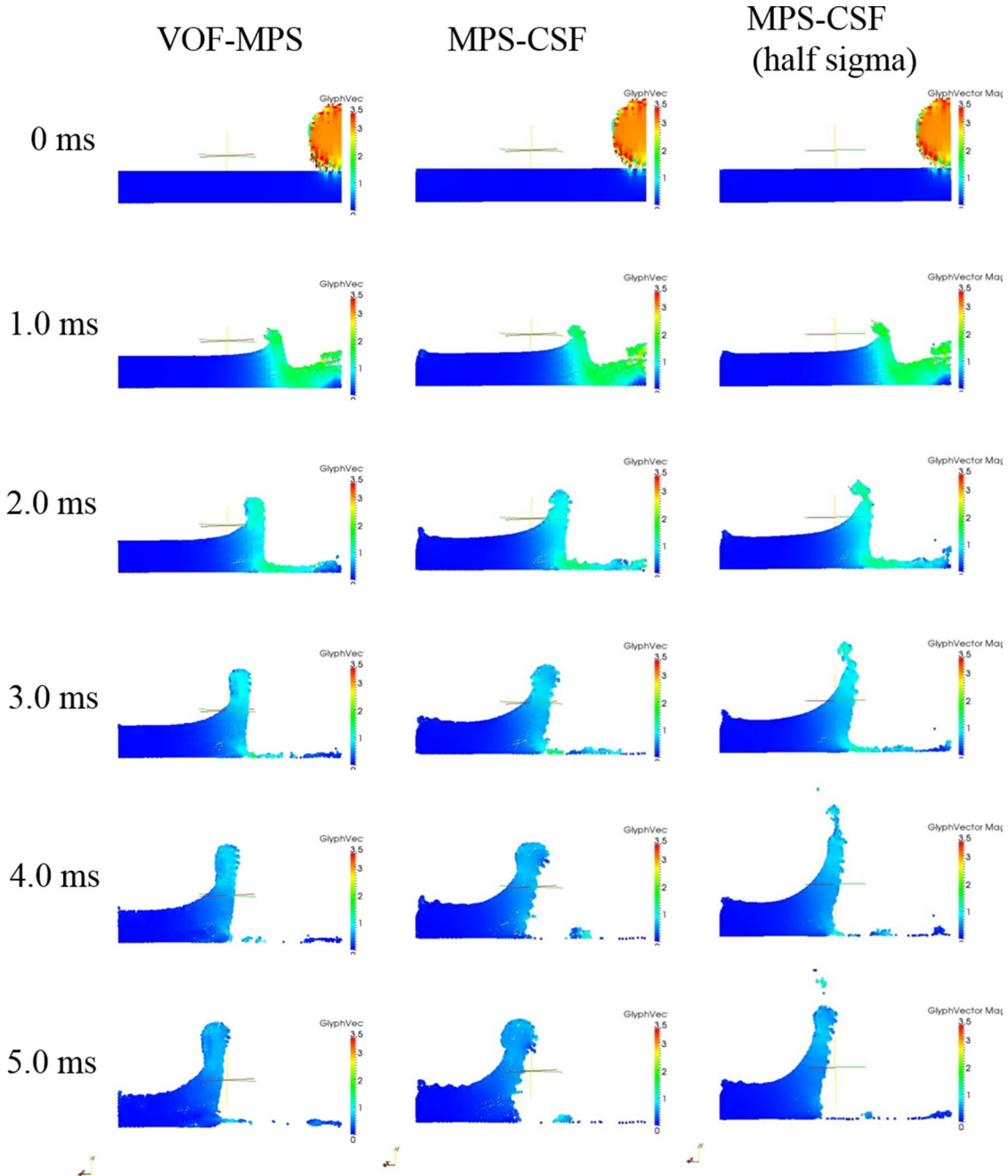


Fig. 9 Time sequential images of simulation results for two different models and reduced surface tension. The particles are colored by the velocity magnitude. The impact velocity is 3.0 m/s

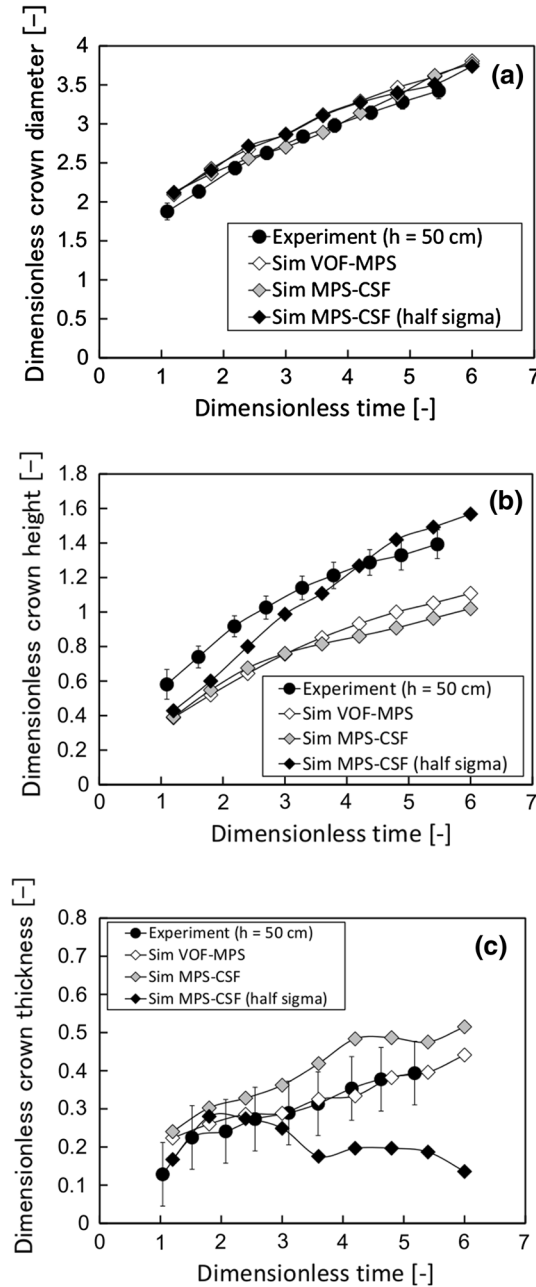


Fig. 10 Comparison of non-dimensional crown parameters. **a** Diameter, **b** height, **c** thickness

interface boundary is highly likely to be originated from the surface tension force calculations employing the interface particles that induce instability along the interface leading to an uneven boundary.

Three plots in Fig. 10 illustrate the temporal variation of crown parameters obtained from the experiment and numerical simulations. Diameter of the crown for all cases is predicted well with slight differences (Fig. 10a). Crown height is underestimated with both methods as it was encountered in the previous section (Fig. 10b). However, VOF-MPS captures the tendency better than the MPS-CSF method. Although the crown height is estimated quantitatively better with the reduced surface tension, the tendency reflecting the rate of change in the height is completely different from the experiment. Moreover, VOF-MPS model is able to estimate the crown thickness sufficiently well, whereas MPS-CSF method overestimates it (Fig. 10c). With the reduced surface tension, MPS-CSF method underestimates the thickness and the results diverge severely after the non-dimensional time of 2.5.

5 Conclusions

Water droplet deposition onto thin water film was investigated with the VOF–MPS method which was developed with the use of OpenFOAM VOF solver. The novel method provided a notable improvement over the former mesh-particle hybrid method with not only introduction of a new surface tension modeling but also the extension to 3D modeling. In addition to the numerical simulation, an experimental study was also performed for the validation of numerical results. Numerical results were validated quantitatively and qualitatively in terms of the crown parameters.

VOF–MPS and MPS–CSF methods demonstrate the similar results from the view point of the tendency. However, MPS–CSF method resolved the liquid–gas interface having more intended structure due to its particle-based surface tension modeling. Evidently, VOF–MPS method provides a remarkable smoothing on the interface and the stability in the solution with the introduction of a new surface tension force modeling to the MPS method.

VOF–MPS method predicted temporal variation of the crown diameter with a slight deviation from the experiments. However, it underestimated the crown height, nevertheless it captured the tendency better than the MPS–CFS method. The finger-like extrusions caused by the interface instabilities led to larger crown height measurements in the experiments. Since both numerical simulations failed to resolve such interface instabilities, underestimations of crown height were observed in the results of both methods.

While MPS–CSF method overestimated the crown thickness, VOF–MPS model demonstrated a notable superiority over the MPS–CSF method in the prediction of crown thickness. On the other hand, both methods could estimate the crown diameter and crown height with almost equivalent level of performance.

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